TABLE 1: Employees

```
mysql> CREATE TABLE Employees (
    ->   emp_id INT PRIMARY KEY,
    ->   emp_name VARCHAR(50),
    ->   dept_id INT,
    ->   salary INT,
    ->   hire_date DATE,
    ->   city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.28 sec)

mysql> desc Employees;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| emp_id    | int         | NO   | PRI | NULL    |       |
| emp_name  | varchar(50) | YES  |     | NULL    |       |
| dept_id   | int         | YES  |     | NULL    |       |
| salary    | int         | YES  |     | NULL    |       |
| hire_date | date        | YES  |     | NULL    |       |
| city      | varchar(50) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
6 rows in set (0.06 sec)
```

TABLE 2 : Projects

```
mysql> CREATE TABLE Projects (
    ->   project_id INT PRIMARY KEY,
    ->   project_name VARCHAR(50),
    ->   dept_id INT,
    ->   budget INT
    -> );
Query OK, 0 rows affected (0.28 sec)

mysql> desc Projects;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| project_id   | int         | NO   | PRI | NULL    |       |
| project_name | varchar(50) | YES  |     | NULL    |       |
| dept_id      | int         | YES  |     | NULL    |       |
| budget       | int         | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
4 rows in set (0.07 sec)
```

TABLE 3 : Attendance

```
mysql> CREATE TABLE Attendance (
    ->   emp_id INT,
    ->   date DATE,
    ->   status VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.47 sec)

mysql> desc Attendance;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| emp_id | int         | YES  |     | NULL    |       |
| date   | date        | YES  |     | NULL    |       |
| status | varchar(20) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
3 rows in set (0.08 sec)
```

TABLE 4: Departments

```
mysql> desc Departments;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| dept_id   | int         | NO   | PRI | NULL    |       |
| dept_name | varchar(50) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
2 rows in set (0.19 sec)
```

SQL QUESTIONS:

Q1:Retrieve all employees.

```
34 •  SELECT * FROM Employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
102	Ravi	2	60000	2021-03-15	Mumbai
103	Neha	1	55000	2020-07-20	Delhi
104	Amit	3	70000	2019-11-01	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
106	Karan	3	72000	2018-08-25	Bangalore
107	Pooja	1	48000	2022-09-30	Delhi
108	Raj	2	52000	2021-12-11	Chennai
109	Meena	3	68000	2020-02-14	Pune
110	Vikas	1	51000	2023-01-18	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q2:Display only emp_name and salary.

```
35 •  SELECT emp_name, salary FROM Employees;
36
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary
Anuj	50000
Ravi	60000
Neha	55000
Amit	70000
Sneha	65000
Karan	72000
Pooja	48000
Raj	52000
Meena	68000
Vikas	51000

Q3:Find employees working in department 1.

```
36 •  SELECT * FROM Employees WHERE dept_id = 1;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
103	Neha	1	55000	2020-07-20	Delhi
107	Pooja	1	48000	2022-09-30	Delhi
110	Vikas	1	51000	2023-01-18	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q4:List employees from Delhi.

```
37 •  SELECT * FROM Employees WHERE city = 'Delhi';
38
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
103	Neha	1	55000	2020-07-20	Delhi
107	Pooja	1	48000	2022-09-30	Delhi
110	Vikas	1	51000	2023-01-18	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q5:Show employees with salary greater than 60000

```
37 •  SELECT * FROM Employees WHERE salary > 60000;
38
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
104	Amit	3	70000	2019-11-01	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
106	Karan	3	72000	2018-08-25	Bangalore
109	Meena	3	68000	2020-02-14	Pune
NULL	NULL	NULL	NULL	NULL	NULL

Q6:Find employees hired after 2021.

```
37 •  SELECT * FROM Employees WHERE hire_date > '2021-12-31';
38
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
105	Sneha	2	65000	2023-05-12	Mumbai
107	Pooja	1	48000	2022-09-30	Delhi
110	Vikas	1	51000	2023-01-18	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q7:Display unique cities.

```
37 •  SELECT DISTINCT city FROM Employees;
38
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city
Delhi
Mumbai
Pune
Bangalore
Chennai

Q8:Sort employees by salary (descending).

```
37 •  SELECT * FROM Employees ORDER BY salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
106	Karan	3	72000	2018-08-25	Bangalore
104	Amit	3	70000	2019-11-01	Pune
109	Meena	3	68000	2020-02-14	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
102	Ravi	2	60000	2021-03-15	Mumbai
103	Neha	1	55000	2020-07-20	Delhi
108	Raj	2	52000	2021-12-11	Chennai
110	Vikas	1	51000	2023-01-18	Delhi
101	Anuj	1	50000	2022-01-10	Delhi
107	Pooja	1	48000	2022-09-30	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q9:Count total employees

```
38 •  SELECT COUNT(*) FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)
10

Q10:Find average salary.

```
39 •  SELECT AVG(salary) FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AVG(salary)
59100.0000

Q11:Find total salary paid per department.

```
40 •  SELECT dept_id, SUM(salary) FROM Employees GROUP BY dept_id;
41
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	SUM(salary)
1	204000
2	177000
3	210000

Q12:Count employees in each department.

```
36  SELECT dept_id, COUNT(*) FROM Employees GROUP BY dept_id;
37
38
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	COUNT(*)
1	4
2	3
3	3

Q13:Find highest salary in each department.

```
36  SELECT dept_id, MAX(salary) FROM Employees GROUP BY dept_id;
37
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	MAX(salary)
1	55000
2	65000
3	72000

Q14:Display employees along with department names.

```sql
SELECT e.emp_name, d.dept_name
FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Anuj	HR
Ravi	Sales
Neha	HR
Amit	IT
Sneha	Sales
Karan	IT
Pooja	HR
Raj	Sales
Meena	IT
Vikas	HR

Q15:List employees working on Sales department

```sql
SELECT e.*
FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id
WHERE d.dept_name = 'Sales';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
102	Ravi	2	60000	2021-03-15	Mumbai
105	Sneha	2	65000	2023-05-12	Mumbai
108	Raj	2	52000	2021-12-11	Chennai

Q16:Find employees whose name starts with 'A'.

```
40 •  SELECT * FROM Employees WHERE emp_name LIKE 'A%';
41
42
43
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
104	Amit	3	70000	2019-11-01	Pune
NULL	NULL	NULL	NULL	NULL	NULL

Q17:Get employees with salary between 50000 and 70000

```
41 •  SELECT * FROM Employees WHERE salary BETWEEN 50000 AND 70000;
42
43
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
102	Ravi	2	60000	2021-03-15	Mumbai
103	Neha	1	55000	2020-07-20	Delhi
104	Amit	3	70000	2019-11-01	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
108	Raj	2	52000	2021-12-11	Chennai
109	Meena	3	68000	2020-02-14	Pune
110	Vikas	1	51000	2023-01-18	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q18:Find employees not from Mumbai.

```
42 •  SELECT * FROM Employees WHERE city != 'Mumbai';
43
44
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
103	Neha	1	55000	2020-07-20	Delhi
104	Amit	3	70000	2019-11-01	Pune
106	Karan	3	72000	2018-08-25	Bangalore
107	Pooja	1	48000	2022-09-30	Delhi
108	Raj	2	52000	2021-12-11	Chennai
109	Meena	3	68000	2020-02-14	Pune
110	Vikas	1	51000	2023-01-18	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q19:Show employees hired in 2022.

```
43 •  SELECT * FROM Employees WHERE YEAR(hire_date) = 2022;
44
45
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

emp_id	emp_name	dept_id	salary	hire_date	city
101	Anuj	1	50000	2022-01-10	Delhi
107	Pooja	1	48000	2022-09-30	Delhi
NULL	NULL	NULL	NULL	NULL	NULL

Q20:Find top 3 highest paid employees.

```
43 • SELECT * FROM Employees ORDER BY salary DESC LIMIT 3;
44
45
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

emp_id	emp_name	dept_id	salary	hire_date	city
106	Karan	3	72000	2018-08-25	Bangalore
104	Amit	3	70000	2019-11-01	Pune
109	Meena	3	68000	2020-02-14	Pune
NULL	NULL	NULL	NULL	NULL	NULL

Q21:INNER JOIN Employees and Departments.

```
43 • SELECT e.emp_name, d.dept_name
44   FROM Employees e
45   INNER JOIN Departments d
46   ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Anuj	HR
Ravi	Sales
Neha	HR
Amit	IT
Sneha	Sales
Karan	IT
Pooja	HR
Raj	Sales
Meena	IT
Vikas	HR

Q22:LEFT JOIN to show all employees with department names.

```sql
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Anuj	HR
Ravi	Sales
Neha	HR
Amit	IT
Sneha	Sales
Karan	IT
Pooja	HR
Raj	Sales
Meena	IT
Vikas	HR

Q23:Find employees working in IT department.

```sql
SELECT e.*
FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id
WHERE d.dept_name = 'IT';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
104	Amit	3	70000	2019-11-01	Pune
106	Karan	3	72000	2018-08-25	Bangalore
109	Meena	3	68000	2020-02-14	Pune

Q24:Show project names with department names.

```sql
SELECT p.project_name, d.dept_name
FROM Projects p
JOIN Departments d ON p.dept_id = d.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

project_name	dept_name
Alpha	HR
Beta	Sales
Gamma	IT
Delta	Sales

Q25:Find employees and their project budgets (via department).

```sql
SELECT e.emp_name, p.budget
FROM Employees e
JOIN Projects p ON e.dept_id = p.dept_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	budget
Anuj	100000
Ravi	120000
Ravi	150000
Neha	100000
Amit	200000
Sneha	120000
Sneha	150000
Karan	200000
Pooja	100000
Raj	120000
Raj	150000
Meena	200000
Vikas	100000

Q26:Find employees earning above average salary.

```sql
SELECT * FROM Employees
WHERE salary > (SELECT AVG(salary) FROM Employees);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary	hire_date	city
102	Ravi	2	60000	2021-03-15	Mumbai
104	Amit	3	70000	2019-11-01	Pune
105	Sneha	2	65000	2023-05-12	Mumbai
106	Karan	3	72000	2018-08-25	Bangalore
109	Meena	3	68000	2020-02-14	Pune
NULL	NULL	NULL	NULL	NULL	NULL

Q27:Rank employees by salary.

```sql
SELECT emp_name, salary,
RANK() OVER (ORDER BY salary DESC) AS rank_no
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	salary	rank_no
Karan	72000	1
Amit	70000	2
Meena	68000	3
Sneha	65000	4
Ravi	60000	5
Neha	55000	6
Raj	52000	7
Vikas	51000	8
Anuj	50000	9
Pooja	48000	10

Q28:Find second highest salary.

```sql
SELECT DISTINCT salary
FROM Employees
ORDER BY salary DESC
LIMIT 1 OFFSET 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

salary
70000

Q29:Get department with highest total salary.

```sql
SELECT dept_id, SUM(salary) AS total
FROM Employees
GROUP BY dept_id
ORDER BY total DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

dept_id	total
3	210000

Q30:Find employees who were absent on 2024-03-01.

26.

```sql
SELECT e.emp_name
FROM Employees e
JOIN Attendance a ON e.emp_id = a.emp_id
WHERE a.status = 'Absent'
AND a.date = '2024-03-01';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name
Ravi
Sneha
Meena